

Ilhan Altunbas

Software Engineer — Generative AI (RAG, Azure), Python, Mobile (Kotlin, KMP)

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Born 30 July 2002 | Driving licence B | Work permit: Belgium | Available immediately

PROFILE

Junior software engineer who builds and runs LLM applications end to end: a bilingual RAG assistant in production on Azure (Azure OpenAI, AI Search, Container Apps), improved through measured changes to latency, cost and answer quality. Coursework in machine learning, computer vision and pattern recognition, a custom-trained YOLO model, and two software internships delivering apps end to end. Uses AI coding tools (Claude Code) daily while keeping code review and measurement in the loop.

SKILLS

Generative AI: LLM integration (Azure OpenAI, Claude), RAG, embeddings, vector search (Azure AI Search, Qdrant), LangChain, LangGraph (tool-calling agent), LLM-as-a-judge evaluation, Langfuse tracing, prompt engineering, query rewriting

Backend: Python, FastAPI, REST APIs, testing (pytest), rate limiting, SQL

Cloud & DevOps: Azure Container Apps, Docker, Docker Compose, GitHub Actions, GitHub Container Registry, Git, Codemagic CI/CD

Machine learning: Computer vision (YOLO object detection), machine learning, pattern recognition

Mobile: Kotlin, Kotlin Multiplatform, Jetpack Compose, MVVM, Coroutines, Room, Ktor

Tools: Claude Code, Android Studio, Postman

Languages: English (Fluent, B2+), French (Intermediate, B1), Turkish (Native)

PROJECTS

DUS Assistant github.com/IlhanAltunbas/DusAssistant

Bilingual (Turkish/English) RAG assistant for the Turkish dental specialty exam, in production on Azure.

- Deployed a FastAPI/LangChain RAG backend to Azure Container Apps with Docker, using Azure OpenAI and Azure AI Search at zero fixed infrastructure cost (scale-to-zero, free tiers).
- Halved answer generation time by profiling latency per step and tuning the model's reasoning effort; cut LLM calls per failing request from 12 to 3 by consolidating retries in the SDK layer.
- Improved follow-up answers by rewriting questions from the chat history before retrieval, and made the answer language reliable (96/96 in repeated tests) by detecting it in code instead of in the prompt.
- Secured the public API against cost abuse with an app key plus per-IP and global daily rate limits; kept providers swappable (Azure OpenAI / Claude, Azure AI Search / Qdrant) behind configuration.
- Built a 28-question bilingual evaluation set with verified source pages, scored with retrieval metrics (hit@5, MRR) and an LLM judge; an agent mode (LangGraph, limits enforced in code) raised fully correct answers from 42% to 72% and cut false refusals from 26% to 4% at comparable faithfulness (77% vs 79%). Backend and mobile tests run in GitHub Actions on every push.

Pedestrian Crossing Violation Detection

Final-year project

- Detected pedestrian-crossing violations automatically by training a custom YOLO (YOLO26) object-detection model, from dataset preparation through training and iteration.

T88Controller github.com/IlhanAltunbas/T88Controller

Kotlin Multiplatform app (Android and iOS) controlling hardware over the network — MrHolo internship.

PROFESSIONAL EXPERIENCE

MrHolo — Software Developer (KMP), Intern

07/2026 – 08/2026

- Delivered T88Controller end to end, from R&D to final product within 2 months, as a single Kotlin Multiplatform codebase for Android and iOS.
- Achieved bidirectional synchronisation with the hardware by translating physical specifications into application logic, using asynchronous TCP sockets and a hexadecimal frame parser with bitwise masking.
- Automated builds for 2 platforms (APK and IPA) by configuring a Codemagic CI/CD pipeline on virtualised macOS.

DivvyDrive — Android Developer, Intern

07/2025 – 08/2025

- Integrated 16 cloud-management REST endpoints with ticket-based authorisation, by designing the MVVM architecture and network layer of 2 Android apps (DivvyDrive, FileReader) with Hilt and Coroutines.
- Enabled large-file uploads by implementing chunked uploads and complex UI state handling.

Çukurova University, Central Library — IT Technician, Part-time

09/2022 – 02/2024

- Developed and maintained the library's websites for 18 months alongside full-time studies, adding multilingual support; troubleshooted peripherals and workstations.

Sezonim — Junior Web Developer (Volunteer), Part-time

10/2021 – 03/2022

- Delivered the responsive front end of the BPWAdana website, a UN Women-funded project, using HTML, CSS, JavaScript and Bootstrap.

EDUCATION

Computer Engineering Degree (4-year programme) — Çukurova University, Turkey

Graduated June 2026

Relevant coursework: Introduction to Machine Learning, Computer Vision, Pattern Recognition